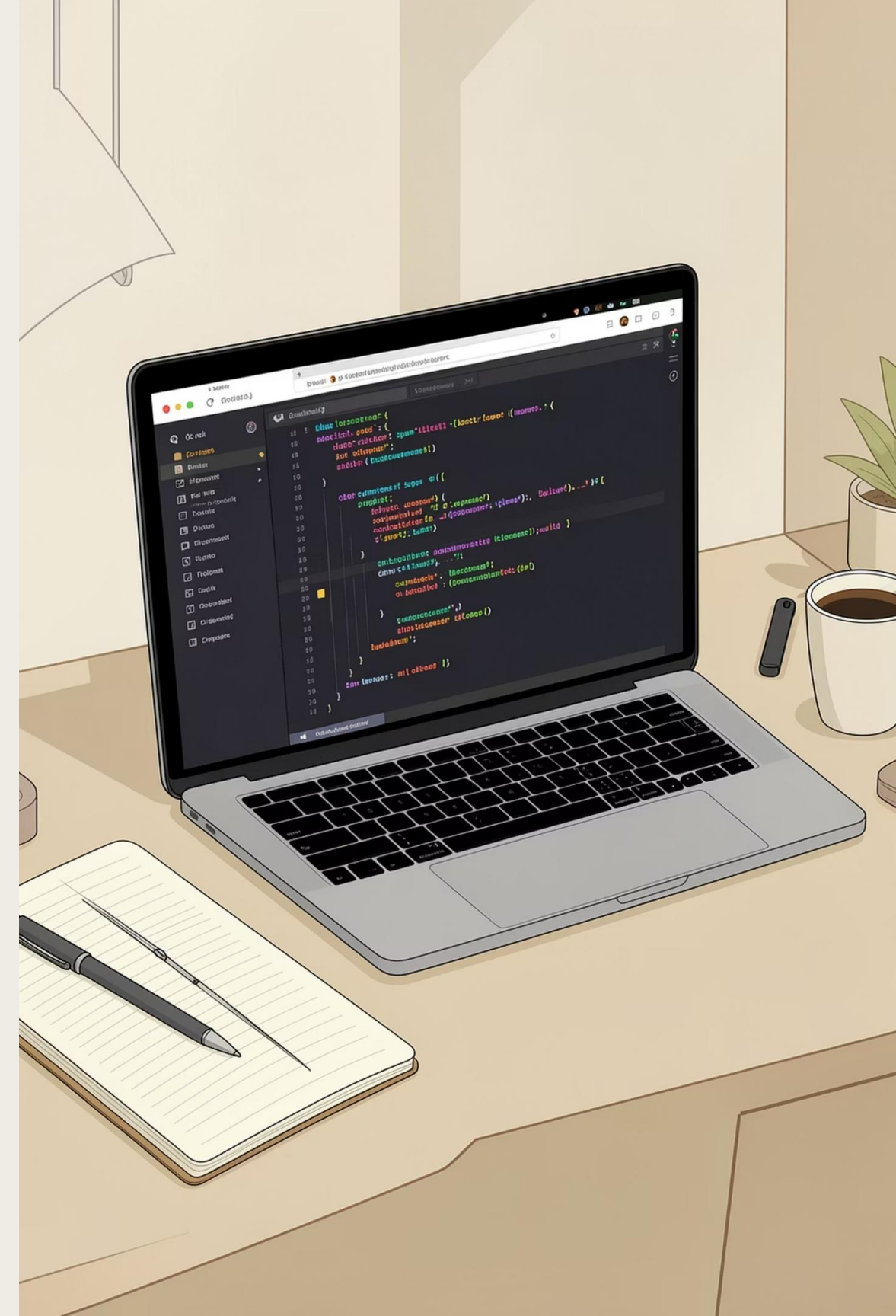


Day 1: HTML & CSS Fundamentals

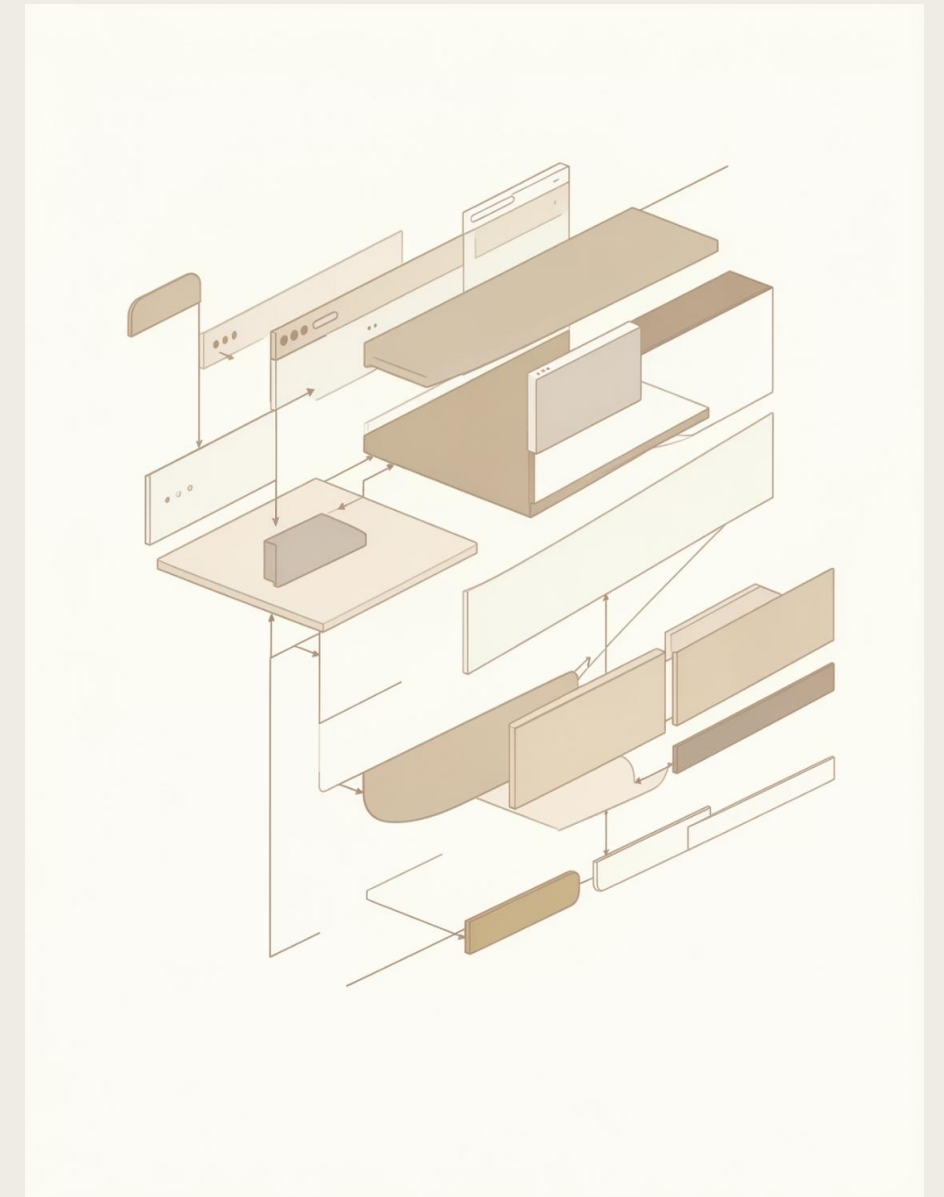
Welcome to your journey into web development! Today, you'll master the foundational building blocks of the web—HTML for structure and CSS for style. By the end of this session, you'll create a complete webpage from scratch.



What is HTML?

HTML (HyperText Markup Language) is the standard language for creating webpages. Think of it as the skeleton of a website—it provides structure and organization for all content.

HTML uses **tags** and **elements** to define different parts of a webpage, from headings and paragraphs to images and links. Every tag has a specific purpose, creating a clear hierarchy.



Anatomy of HTML Tags



Opening Tag

Starts with `<` and ends with `>`.
Contains the element name.



Content

The actual text, image, or data that
appears on your webpage.



Closing Tag

Begins with `</` and ends with `>`.
Completes the element.

Some tags are **self-closing** (like `<img>` or `<br>`) and don't need separate closing tags.

Basic Document Structure

01

<html>

Root element—tells the browser this is an HTML document

02

<head>

Contains metadata, title, and links to stylesheets

03

<body>

Holds all visible content on your webpage



```
1 /html
2 <html
3   </html>
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5   <head> body
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7   </head>
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9   <body>
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11   </body>
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13 </html>
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<i>Chlecafī lqrfīsal</i>	<i>Hdleud-tiiregemure</i>
<i>Cmdagies</i>	<i>Ohfjrwga, cruae</i>
<i>Qvotemg^f cboan</i>	<i><u>Engobtalmgē axmxy</u></i>
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<i>Coistoni; Bd:</i>	<i>—————</i>
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Working with Text

Headings

Use `<h1>` through `<h6>` for different levels of importance. `<h1>` is largest, `<h6>` is smallest.

```
<h1>Main Heading</h1>
<h2>Subheading</h2>
<h3>Section Title</h3>
```

Paragraphs & Formatting

`<p>` creates paragraphs. Use `<b>` for **bold**, `<i>` for *italics*, and `<br>` for line breaks.

```
<p>This is a <b>bold</b>
paragraph
with a line
break<br>right here.</p>
```

Links & Images

1

Creating Links

The `<a>` tag creates clickable links. Use the `href` attribute to specify the destination.

```
<a href="https://example.com">  
  Click here  
</a>
```

2

Adding Images

The `<img>` tag embeds images. Requires `src` (source) and `alt` (description) attributes.

```

```

Working with Lists

Ordered Lists

Numbered lists using `<ol>` and `<li>` tags.

```
<ol>
  <li>First item</li>
  <li>Second item</li>
  <li>Third item</li>
</ol>
```

Unordered Lists

Bullet-point lists using `<ul>` and `<li>` tags.

```
<ul>
  <li>Apples</li>
  <li>Oranges</li>
  <li>Bananas</li>
</ul>
```




Hands-On Task



Create Your HTML File

Set up the basic structure with `<html>`, `<head>`, and `<body>` tags



Add Content

Include at least one heading, two paragraphs, and a list



Insert Elements

Add a link to your favorite website and an image



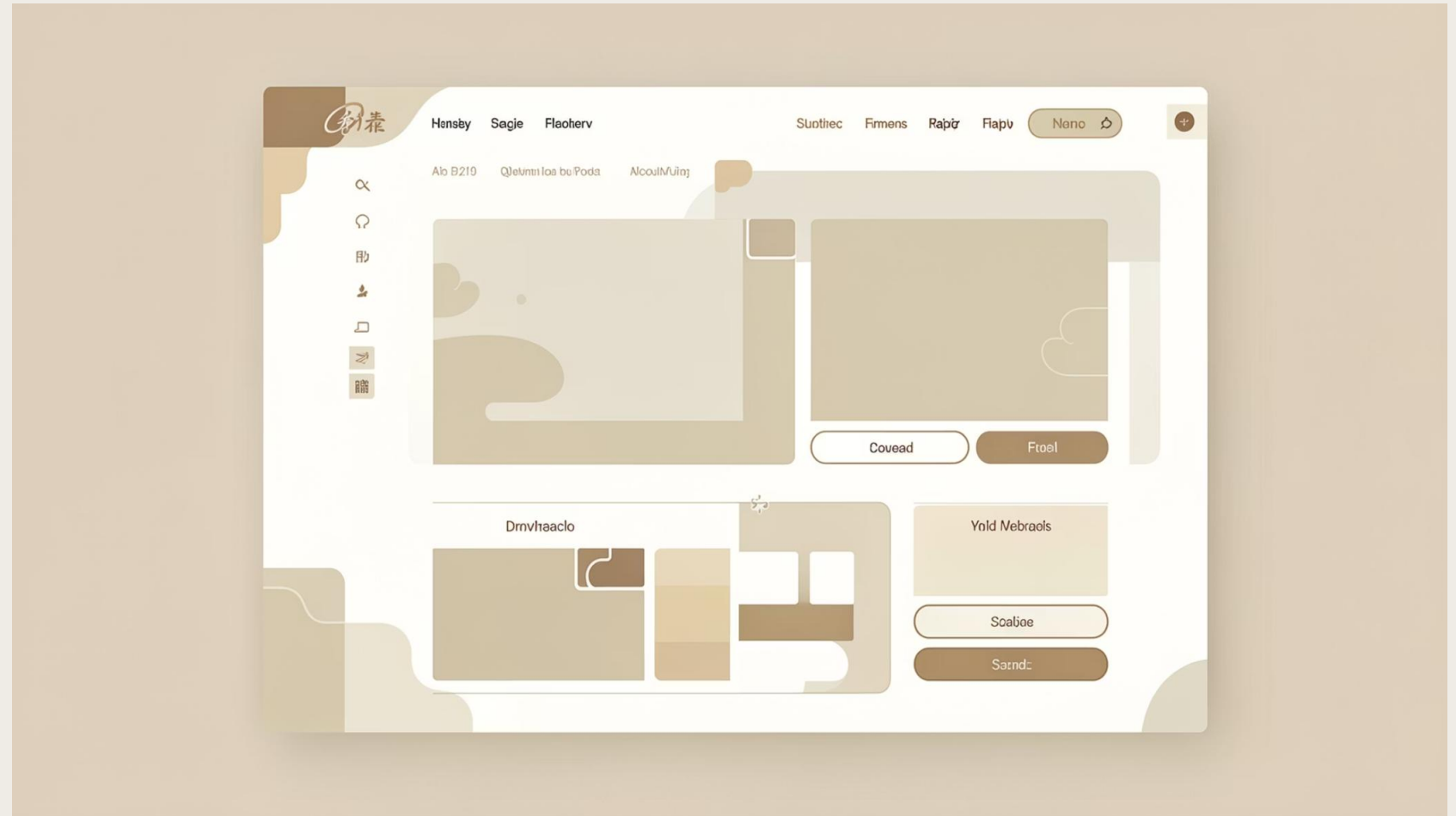
Test Your Page

Open the file in a browser and verify everything displays correctly

What is CSS?

CSS (Cascading Style Sheets) controls the presentation of your HTML—colors, fonts, spacing, and layout. While HTML builds the structure, CSS adds the visual polish that makes websites beautiful.

CSS uses **selectors** to target HTML elements and **properties** to define their appearance. This separation keeps your code clean and maintainable.



Colors & Backgrounds

Set text and background colors with color names or hex codes



Spacing

Control margins (outside) and padding (inside) to create breathing room



Layout

Position elements using tools like Flexbox for responsive designs

CSS Integration Methods

Inline CSS

Add styles directly to HTML elements using the `style` attribute. Quick but hard to maintain.

```
<h1 style="color: blue">  
  Hello World  
</h1>
```

Internal CSS

Place styles in a `<style>` block within the `<head>` section. Better for single pages.

```
<style>  
  h1 { color: blue; }  
</style>
```

External CSS

Link to a separate `.css` file. Best practice—keeps code organized and reusable across multiple pages.

```
<link rel="stylesheet"  
      href="styles.css">
```

Your CSS Challenge

Take the HTML page you created earlier and enhance it with CSS! Add colors to your text and background, adjust spacing with margins and padding, and experiment with different fonts. Use external CSS for the cleanest approach.